WAYHUNTER ERD v1.1 — SOURCE & SCAN EXTENSION (PROPOSED)

Controlled review | 10 November 2026. The attached original ERD has **118 baseline proposed tables**, unchanged. This supplement proposes a voucher-source table and an **optional** OCR scan candidate table; it does not alter historical schema or claim production approval.

Voucher intake traceability (ADM-12, new working process p.13)

Entity	FK references	Description/status
`PartnerOffers`	`PartnerId`	EXISTING offer and actual agreement conditions
`VoucherInventoryItems`	`PartnerOfferId`	EXISTING unique available stock, code hash, encrypted ref, expiry
`MediaAssets`	`OwnerUserId`	EXISTING private image/PDF metadata
`VoucherSourceDocuments`	Offer, InventoryItem?, MediaAsset?, UploaderUser, VerifierUser	**PROPOSED** upload/manual-entry provenance + verify state
`VoucherScanResults`	VoucherSourceDocument, ReviewerUser?	**OPTIONAL / NOT APPROVED** OCR/QR/ barcode candidates, no raw code
`AdminReviewActions` / `AdminAuditEvents`	Case, Actor	EXISTING review history, reason and authorization

Relationship map

```
PartnerOffers (1) ----- (M) VoucherInventoryItems [existing]
      |                          |
      +---- (M) VoucherSourceDocuments (0..1) -> MediaAssets [private]
                        |     |
                        |     +--> Users [uploaded by / verified by]
                        +---- (M) VoucherScanResults [OPTIONAL; reviewed by Users]

RedemptionRequests (1) -- (0..M) CoinReservations [existing]
          |
          +-- VoucherAssignments -> VoucherInventoryItems -> VoucherUses
```

VCH-DEC-01 financial trigger (BLOCKER)

New Admin Working Process p.14 requires final debit at Fulfillment; BRD v2.0 requires final debit on Admin Approval. **Neither has been silently chosen.** `CoinReservations` hold pending Coins; `CoinLedgerEntries`

final debit once after owner-approved trigger; reject/cancel releases once. Unique assignment and expiry audit continue independently.

Migration status

`Wayhunter_SQL_Voucher_Extension_PROPOSED.sql` creates only a proposed SourceDocuments table if someone explicitly executes it in a review database. OCR table is **commented out**. No automatic database changes performed. The editable Draw.io keeps 13 original diagrams and adds a separate proposed page `14 Voucher v1.1 Proposed`.

WAYHUNTER

FULL ENTITY RELATIONSHIP DIAGRAM

Mobile App + Angular Admin + ASP.NET Core + SQL Server

118	200	13	17
Proposed SQL Tables	Foreign Key Links	Database Domains	Admin Modules

DESIGN BASIS

Master BRD v2.0 (24 pages) • Backend Architecture v3.0 (16 pages) • Flutter Architecture v2.0 (13 pages)

Scope: all user/mobile workflows, 17 Admin Portal modules, Coins-only wallet, map/places, safety and integrations.

All table/column/relationship names are PROPOSED engineering design, not extracted from a live database.

BRD is a DRAFT awaiting formal approval. Conditional tables and unresolved decisions are explicitly identified.

PACKAGE CONTENTS

13 domain relationship diagrams + high-level overview + Admin mapping + critical constraints.

Accompanying editable draw.io, Mermaid, SQL Server DDL, and complete field-by-field data dictionary.

For review and implementation planning. Confirm pending product decisions before production migrations.

ADMIN PORTAL: ALL 17 MODULES

Verified against BRD section 11 (pp.15-17) and Backend Architecture section 09 (p.10)

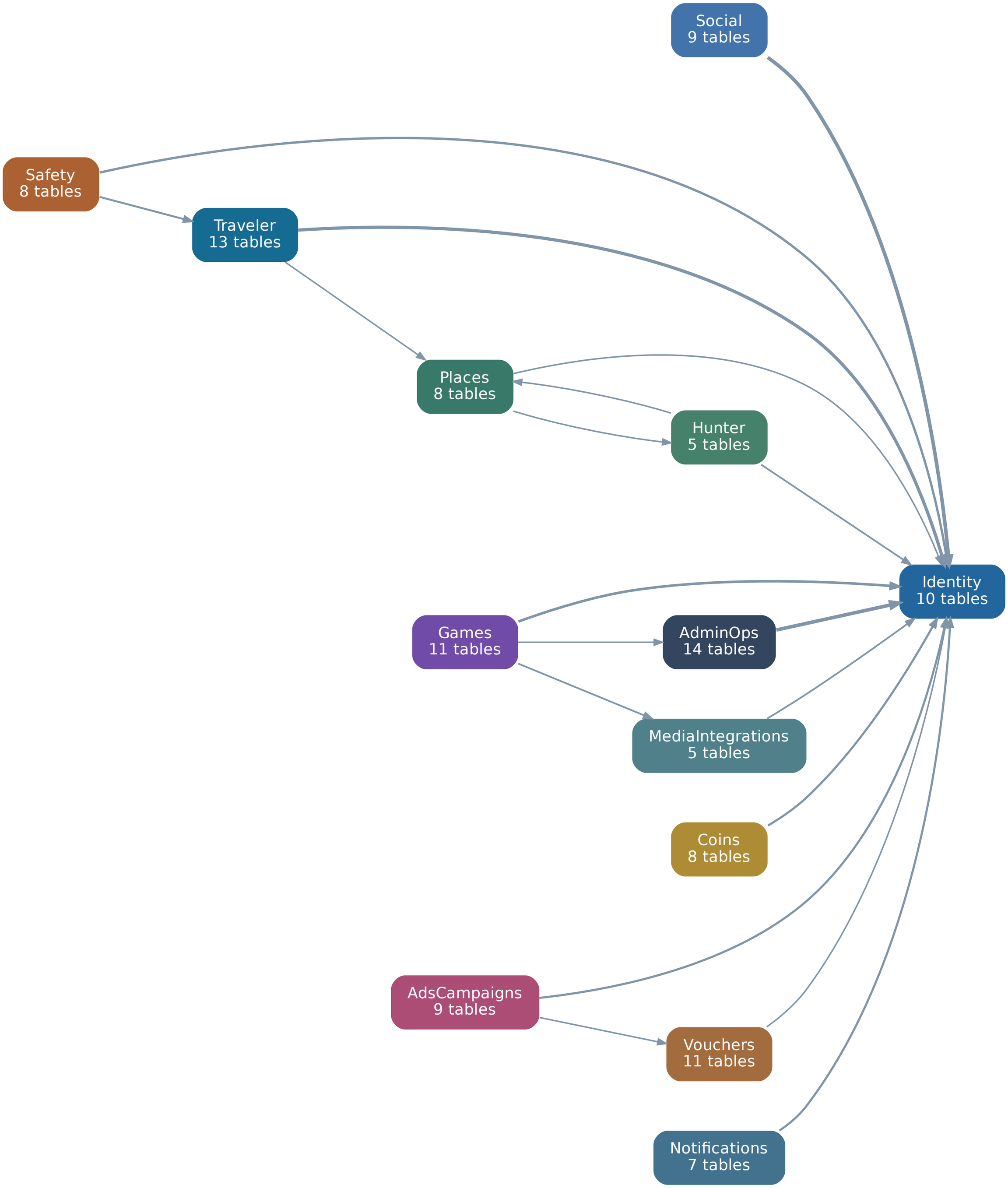
Every required Admin module is represented by relational base data or derived queries.

ID	ADMIN MODULE	LINKED TABLES / DATA SOURCES	TYPE
ADM-01	Admin Login	Users, UserSessions, AdminUserRoles, AdminRoles, AdminPermissions, AdminAuditEvents	Verified
ADM-02	Dashboard	Derived from Users, Trips, HunterSubmissions, Games, CoinLedgerEntries, RedemptionRequests, Advertisements	Derived / no dedicated Dashboard table
ADM-03	User Management	Users, UserProfiles, UserDevices, AccountModerationActions, UserConsents, AdminAuditEvents	Verified
ADM-04	Hunter Verification	HunterSubmissions, HunterEvidence, HunterDuplicateCandidates, HunterReviews, HunterRiskFlags, AdminReviewCases	Verified
ADM-05	Advertisement Management	Advertisements, AdPlacements, AdPlacementAssignments, AdDistrictTargets, AdImpressions, AdClicks	Verified
ADM-06	Places Management	Places, PlaceMedia, PlaceHistory, PlaceMerges, PlaceSourceRecords	Verified
ADM-07	Trips & Historical Routes	Trips, TripStops, TripSessions, HistoricalRouteSegments, TripRoutePlans	Verified
ADM-08	Hazards	HazardReports, HazardEvidence, HazardReviews, HazardAlerts, RestrictedZones	Verified
ADM-09	Group Trips	TripMembers, TripInvitations, TripShareConsents, TripLocationSamples, TripSafetyCheckins, SafetyIncidents	Verified
ADM-10	Game Management & Anti-cheat	Games, GameStages, GamePublicationReviews, GameAttempts, GameStageProofs, GameCompletionReviews, GameAntiCheatFlags, GameExpiryEvents	Verified
ADM-11	Coins & Rewards	CoinWallets, CoinLedgerEntries, CoinRewardSources, CoinReservations, CoinAdjustmentRequests, CoinRuleVersions	Verified
ADM-12	Merchant / Partner Management	Partners, PartnerContracts, PartnerBranches, PartnerOffers, VoucherInventoryItems	Verified
ADM-13	Redemptions / Settlements	RedemptionRequests, CoinReservations, VoucherAssignments, VoucherUses, PartnerSettlements, SettlementLines	Verified; payout conditional
ADM-14	Campaign Management	Campaigns, CampaignAssignments, CampaignParticipations, Advertisements, AdDistrictTargets	Verified
ADM-15	Reports & Analytics	ReportExports, IntegrationUsageRecords, CoinLedgerEntries, Trips, Games, Ads / derived metrics	Verified; aggregate views derived
ADM-16	Notifications	NotificationTemplates, Notifications, NotificationDeliveries, NotificationPreferences, ScheduledReminders, OutboxMessages	Verified
ADM-17	Settings, Restricted Zones, Audit & Admin Mgmt	SystemRuleSettings, SystemRuleVersions, RestrictedZones, AdminRoles, AdminPermissions, AdminUserRoles, AdminAuditEvents, FeatureFlags	Verified

Note: Dashboard, three Leaderboards, and most KPIs are derived read models - not duplicate editable base tables.

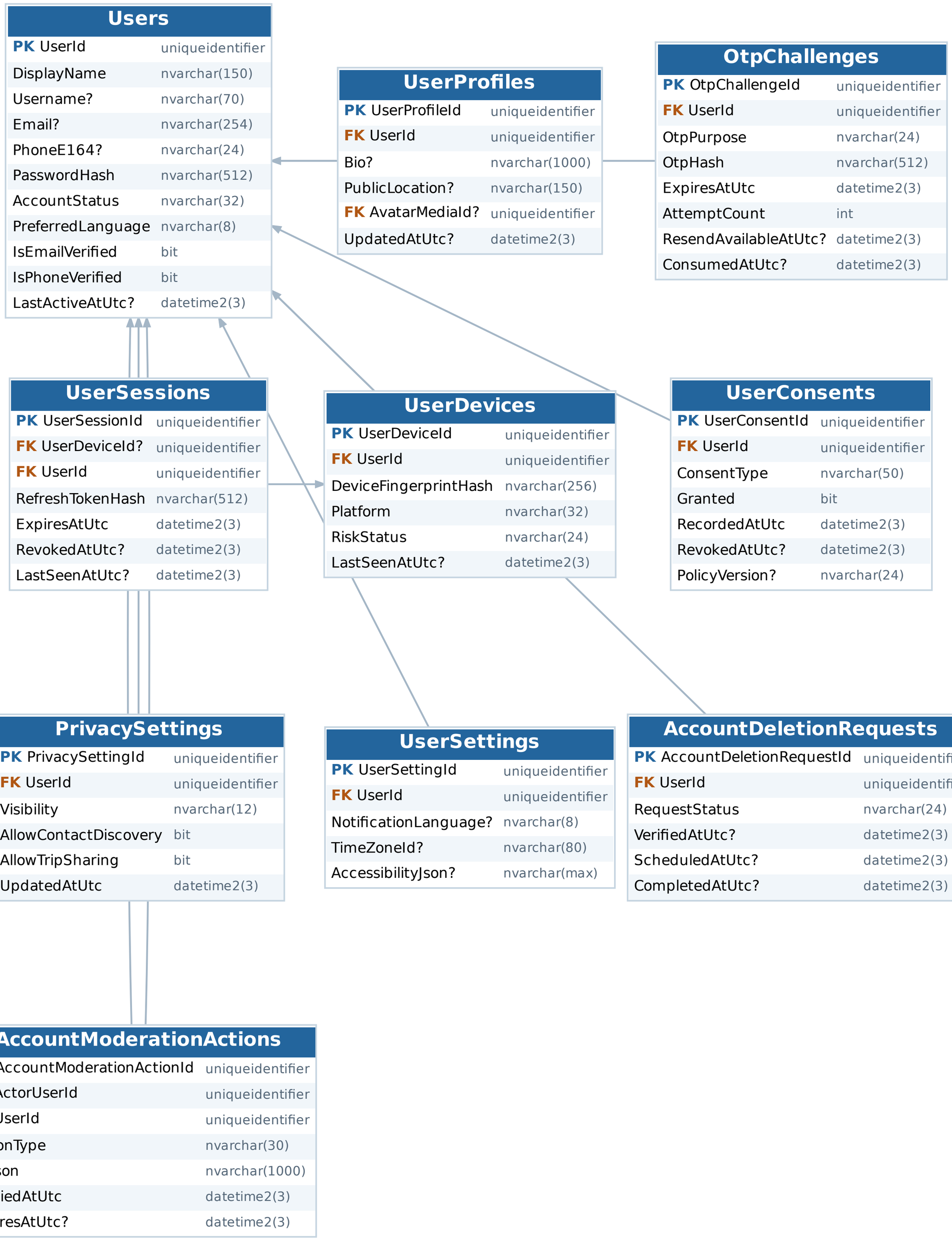
DATA DOMAIN OVERVIEW

Cross-domain dependencies. Each table, field and FK appears on domain pages and in the CSV dictionary.



01 / IDENTITY - 10 TABLES

User registration, OTP, session, consent, privacy | FK arrows: child N -> parent 1



02 / SOCIAL - 9 TABLES

Friends, contacts opt-in, social profile content | FK arrows: child N -> parent 1

Follows	
PK FollowId	uniqueidentifier
FK FollowedUserId	uniqueidentifier
FK FollowerUserId	uniqueidentifier
FollowedAtUtc	datetime2(3)

FollowRequests	
PK FollowRequestId	uniqueidentifier
FK TargetUserId	uniqueidentifier
FK RequesterUserId	uniqueidentifier
RequestStatus	nvarchar(24)
RespondedAtUtc?	datetime2(3)

UserBlocks	
PK UserBlockId	uniqueidentifier
FK BlockedUserId	uniqueidentifier
FK BlockerUserId	uniqueidentifier
Reason?	nvarchar(500)
BlockedAtUtc	datetime2(3)

ContactDiscoveryConsents	
PK ContactDiscoveryConsentId	uniqueidentifier
FK UserId	uniqueidentifier
IsOptedIn	bit
UpdatedAtUtc	datetime2(3)
ProofVersion?	nvarchar(24)

ContactMatchEvents	
PK ContactMatchEventId	uniqueidentifier
FK MatchedUserId	uniqueidentifier
FK RequesterUserId	uniqueidentifier
MatchedAtUtc	datetime2(3)
Method	nvarchar(24)
DisclosedToUser	bit

FriendShareInvites	
PK FriendShareInviteId	uniqueidentifier
FK InviterUserId	uniqueidentifier
InviteTokenHash	nvarchar(128)
Channel?	nvarchar(30)
ExpiresAtUtc?	datetime2(3)
AcceptedAtUtc?	datetime2(3)

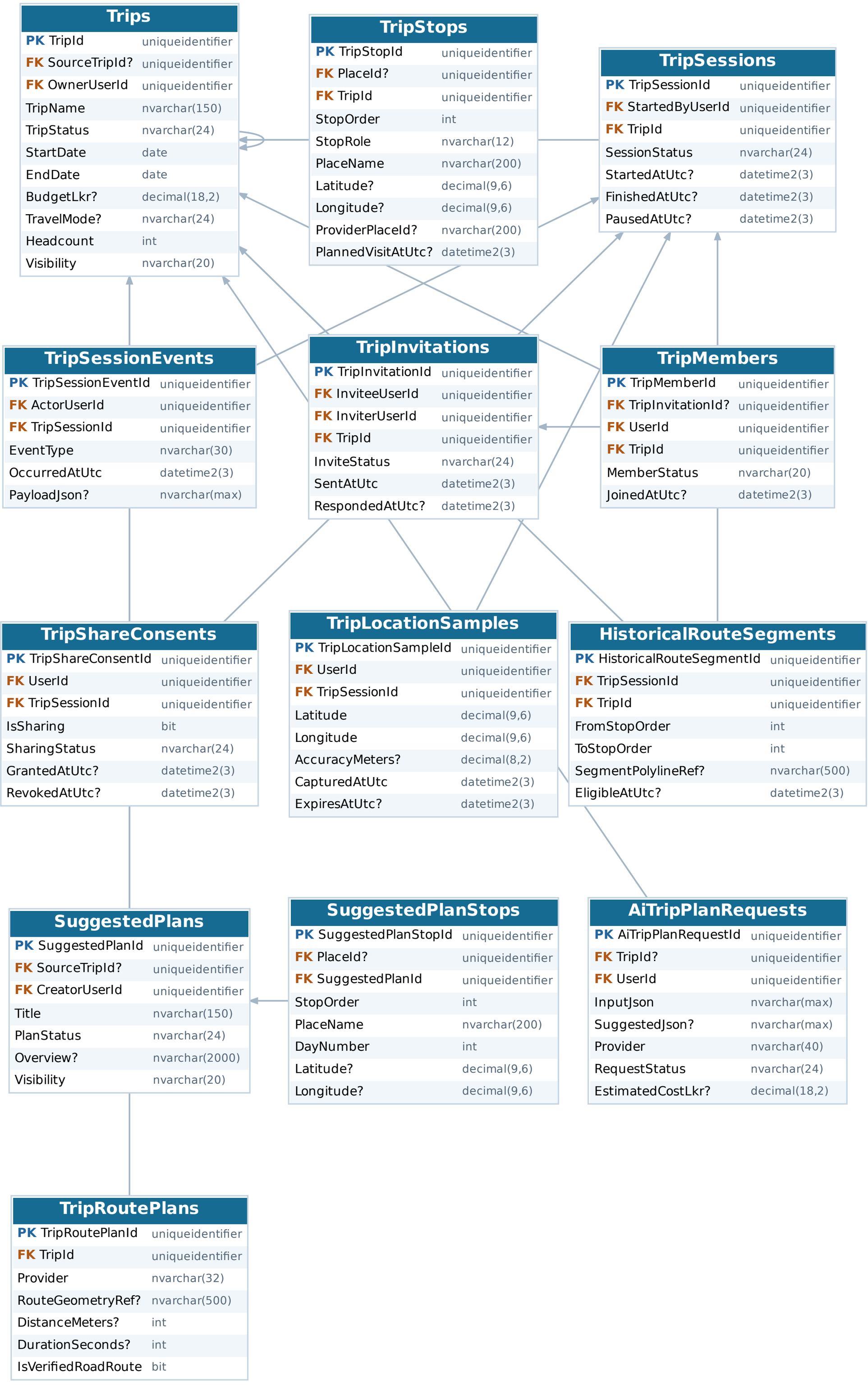
UserPosts	
PK UserPostId	uniqueidentifier
FK MediaAssetId?	uniqueidentifier
FK PlaceId?	uniqueidentifier
FK AuthorUserId	uniqueidentifier
Caption?	nvarchar(2000)
Visibility	nvarchar(20)
PostStatus	nvarchar(24)

PostComments	
PK PostCommentId	uniqueidentifier
FK AuthorUserId	uniqueidentifier
FK UserPostId	uniqueidentifier
Body	nvarchar(1000)
Status	nvarchar(20)

PostReactions	
PK PostReactionId	uniqueidentifier
FK UserId	uniqueidentifier
FK UserPostId	uniqueidentifier
ReactionType	nvarchar(20)

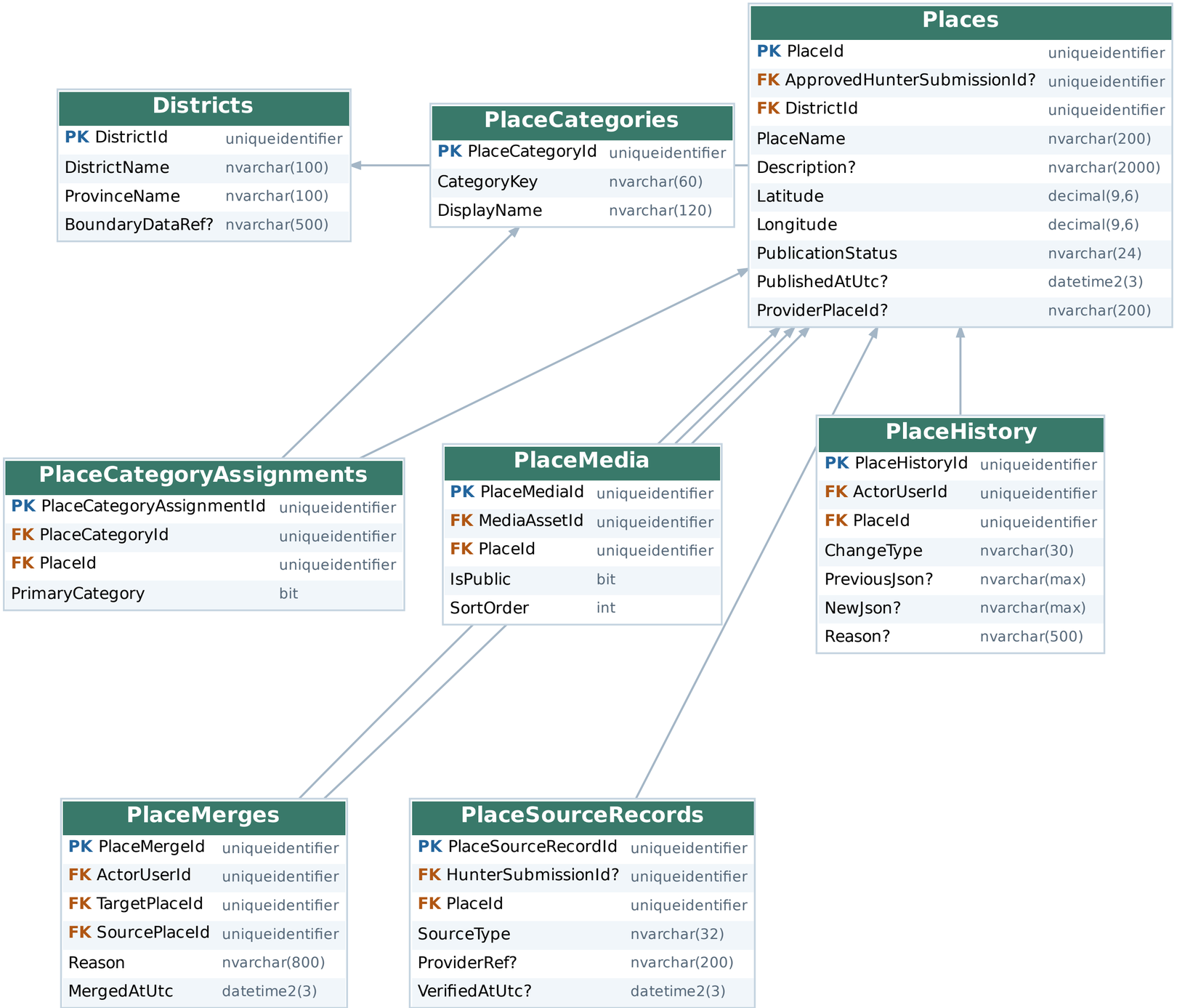
03 / TRAVELER - 13 TABLES

Trip plans, stops, sessions, sharing, historical routes | FK arrows: child N -> parent 1



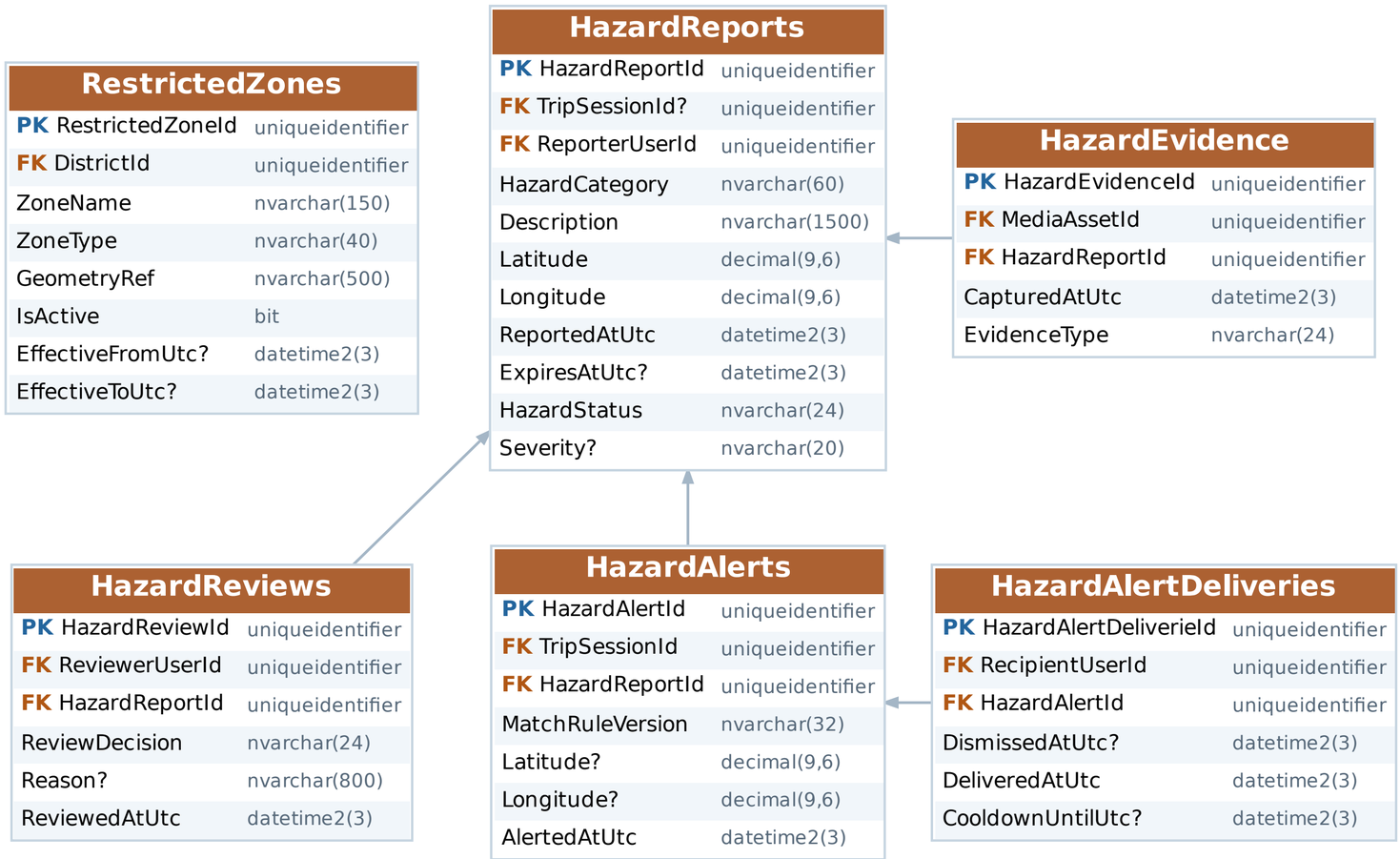
04 / PLACES - 8 TABLES

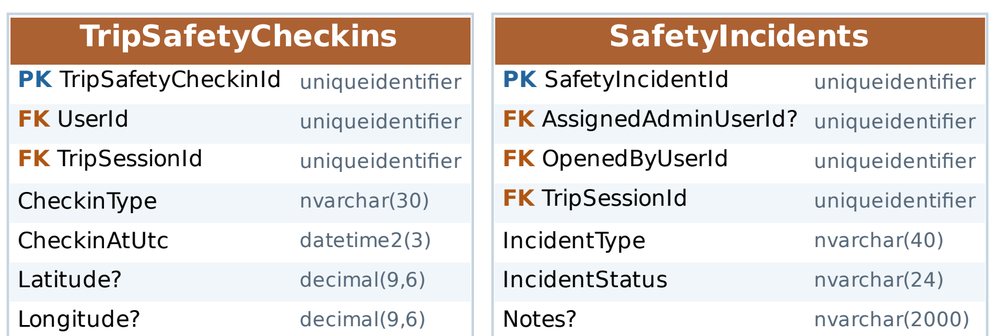
Approved place catalogue, districts, provenance | FK arrows: child N -> parent 1



05 / SAFETY - 8 TABLES

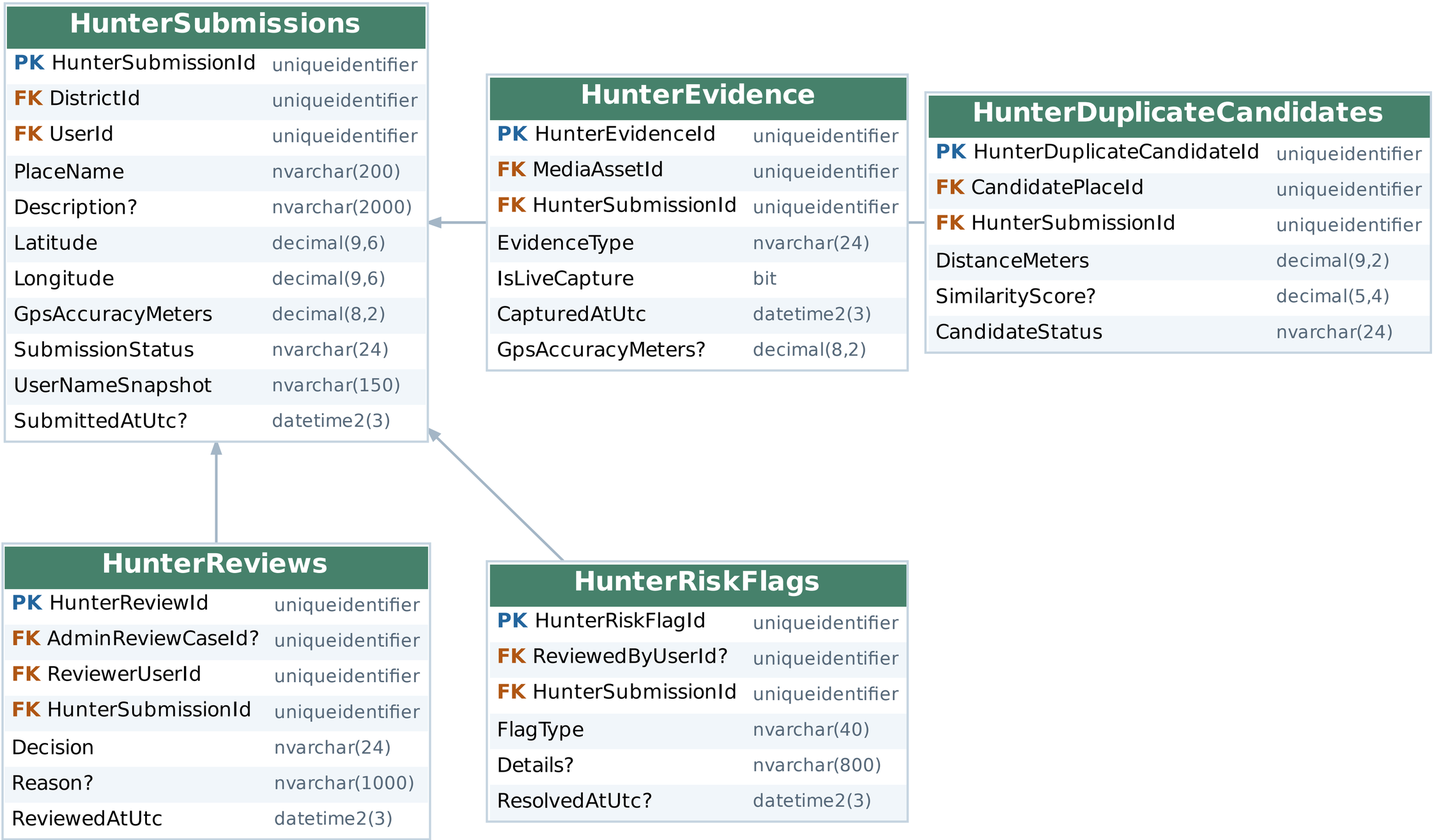
Hazards, route popups, restricted zones and safety | FK arrows: child N -> parent 1





06 / HUNTER - 5 TABLES

Live evidence, duplicate screening, Admin reviews | FK arrows: child N -> parent 1



07 / GAMES - 11 TABLES

Game creation/attempts, proof verification, expiry | FK arrows: child N -> parent 1

Games

PK GameId	uniqueidentifier
FK CampaignId?	uniqueidentifier
FK CoverMediaId?	uniqueidentifier
FK CreatorUserId	uniqueidentifier
Title	nvarchar(150)
Description?	nvarchar(2000)
CreationStatus	nvarchar(24)
PublishedAtUtc?	datetime2(3)
ExpiresAtUtc?	datetime2(3)
ArchivedAtUtc?	datetime2(3)
DurationSeconds?	int
UserNameSnapshot	nvarchar(150)

GameStages

PK GameStageId	uniqueidentifier
FK GameId	uniqueidentifier
StageOrder	int
Title?	nvarchar(150)
TimeLimitSeconds?	int

GameClues

PK GameClueId	uniqueidentifier
FK MediaAssetId?	uniqueidentifier
FK GameStageId	uniqueidentifier
ClueType	nvarchar(20)
ClueText?	nvarchar(1500)
SortOrder	int

GamePublicationReviews

PK GamePublicationReviewId	uniqueidentifier
FK AdminReviewCaseId?	uniqueidentifier
FK ReviewerUserId	uniqueidentifier
FK GameId	uniqueidentifier
Decision	nvarchar(24)
Reason?	nvarchar(1000)
ReviewedAtUtc	datetime2(3)

GameAttempts

PK GameAttemptId	uniqueidentifier
FK PlayerUserId	uniqueidentifier
FK GameId	uniqueidentifier
AttemptStatus	nvarchar(24)
StartedAtUtc	datetime2(3)
FinishedAtUtc?	datetime2(3)
ElapsedSeconds?	int

GameStageProofs

PK GameStageProofId	uniqueidentifier
FK MediaAssetId	uniqueidentifier
FK GameStageId	uniqueidentifier
FK GameAttemptId	uniqueidentifier
ProofStatus	nvarchar(24)
CapturedAtUtc	datetime2(3)
IsLiveCapture	bit

GameStageProofReviews

PK GameStageProofReviewId	uniqueidentifier
FK ReviewerUserId	uniqueidentifier
FK GameStageProofId	uniqueidentifier
Decision	nvarchar(24)
Reason?	nvarchar(1000)
ReviewedAtUtc	datetime2(3)

GameCompletionReviews

PK GameCompletionReviewId	uniqueidentifier
FK AdminReviewCaseId?	uniqueidentifier
FK ReviewerUserId	uniqueidentifier
FK GameAttemptId	uniqueidentifier
Decision	nvarchar(24)
Reason?	nvarchar(1000)
ReviewedAtUtc	datetime2(3)

GameReports

PK GameReportId	uniqueidentifier
FK ReporterUserId	uniqueidentifier
FK GameId	uniqueidentifier
ReportType	nvarchar(40)
ReportStatus	nvarchar(24)
Reason	nvarchar(1000)

GameAntiCheatFlags

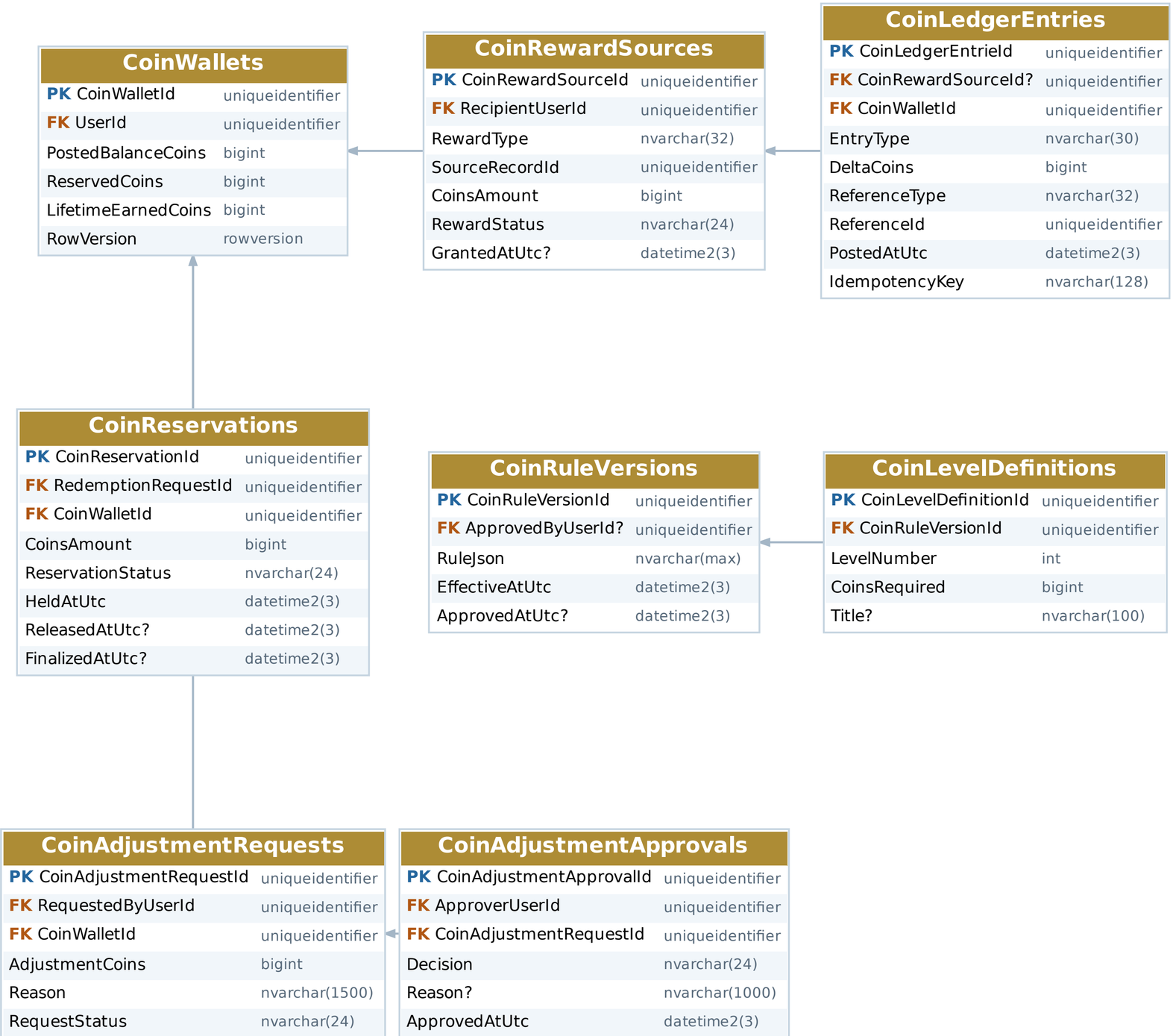
PK GameAntiCheatFlagId	uniqueidentifier
FK ActorUserId?	uniqueidentifier
FK GameAttemptId?	uniqueidentifier
FK GameId	uniqueidentifier
FlagType	nvarchar(40)
Reason	nvarchar(1000)
Disposition?	nvarchar(30)

GameExpiryEvents

PK GameExpiryEventId	uniqueidentifier
FK GameId	uniqueidentifier
ExpiredAtUtc	datetime2(3)
ExpiryRuleVersion	nvarchar(24)
EventStatus	nvarchar(20)

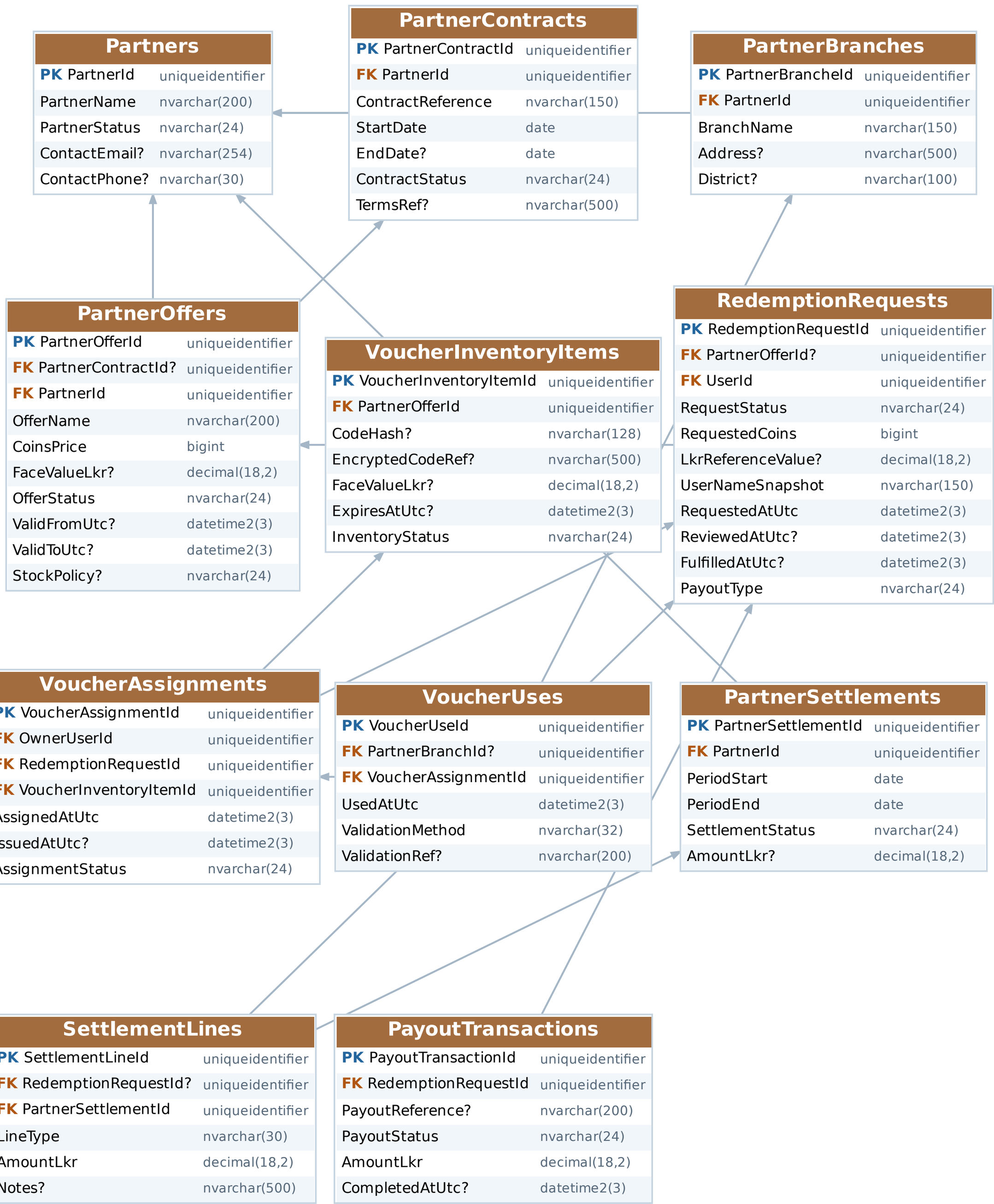
08 / COINS - 8 TABLES

Coins-only wallets, verified earnings, reservations | FK arrows: child N -> parent 1



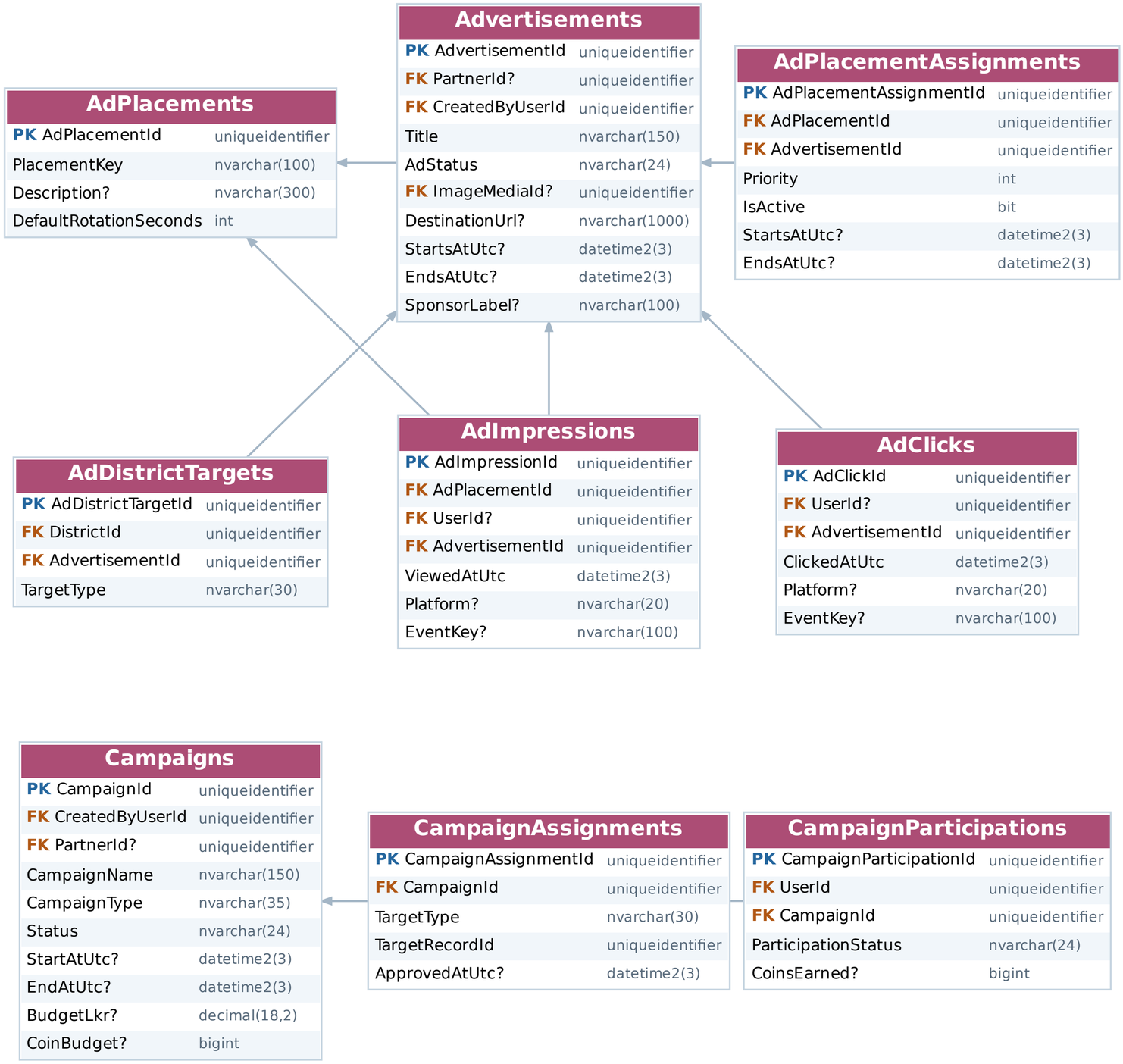
09 / VOUCHERS - 11 TABLES

Partner contracts, redemptions, voucher issue and use | FK arrows: child N -> parent 1



10 / ADSCAMPAIGNS - 9 TABLES

Ads, district targeting, campaigns and budget | FK arrows: child N -> parent 1



11 / NOTIFICATIONS - 7 TABLES

Push/in-app notifications, reminders and outbox | FK arrows: child N -> parent 1

NotificationTemplates

PK NotificationTemplateId	uniqueidentifier
TemplateKey	nvarchar(100)
Language	nvarchar(8)
TitleText	nvarchar(200)
BodyText	nvarchar(1000)
IsActive	bit

Notifications

PK NotificationId	uniqueidentifier
FK NotificationTemplateId?	uniqueidentifier
FK RecipientUserId	uniqueidentifier
NotificationType	nvarchar(40)
Title	nvarchar(200)
Body	nvarchar(1000)
ReferenceType?	nvarchar(30)
ReferenceId?	uniqueidentifier
IsRead	bit
CreatedForAtUtc	datetime2(3)

NotificationDeliveries

PK NotificationDelivereId	uniqueidentifier
FK NotificationId	uniqueidentifier
DeliveryChannel	nvarchar(24)
DeliveryStatus	nvarchar(24)
AttemptCount	int
LastAttemptAtUtc?	datetime2(3)
ProviderMessageId?	nvarchar(200)

NotificationPreferences

PK NotificationPreferenceId	uniqueidentifier
FK UserId	uniqueidentifier
Category	nvarchar(40)
PushEnabled	bit
InAppEnabled	bit

UserPushTokens

PK UserPushTokenId	uniqueidentifier
FK UserDeviceId?	uniqueidentifier
FK UserId	uniqueidentifier
TokenCiphertextRef	nvarchar(500)
Platform	nvarchar(20)
IsActive	bit
LastSeenAtUtc?	datetime2(3)

ScheduledReminders

PK ScheduledReminderId	uniqueidentifier
FK RedemptionRequestId?	uniqueidentifier
FK TripId?	uniqueidentifier
FK UserId	uniqueidentifier
ReminderType	nvarchar(40)
DueAtUtc	datetime2(3)
ReminderStatus	nvarchar(24)
DeduplicationKey	nvarchar(128)

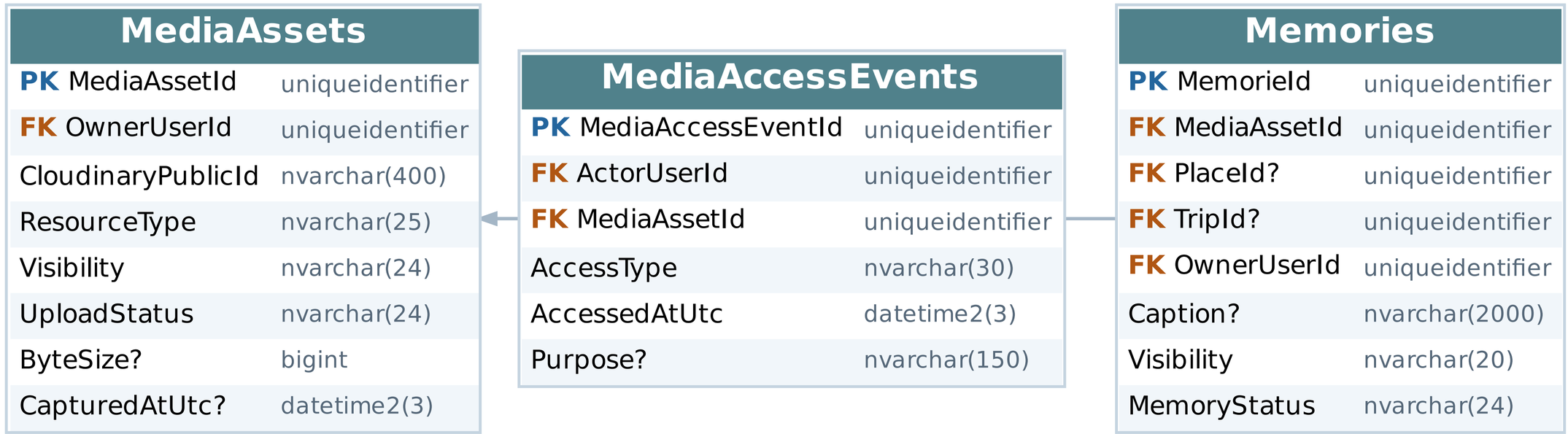
OutboxMessages

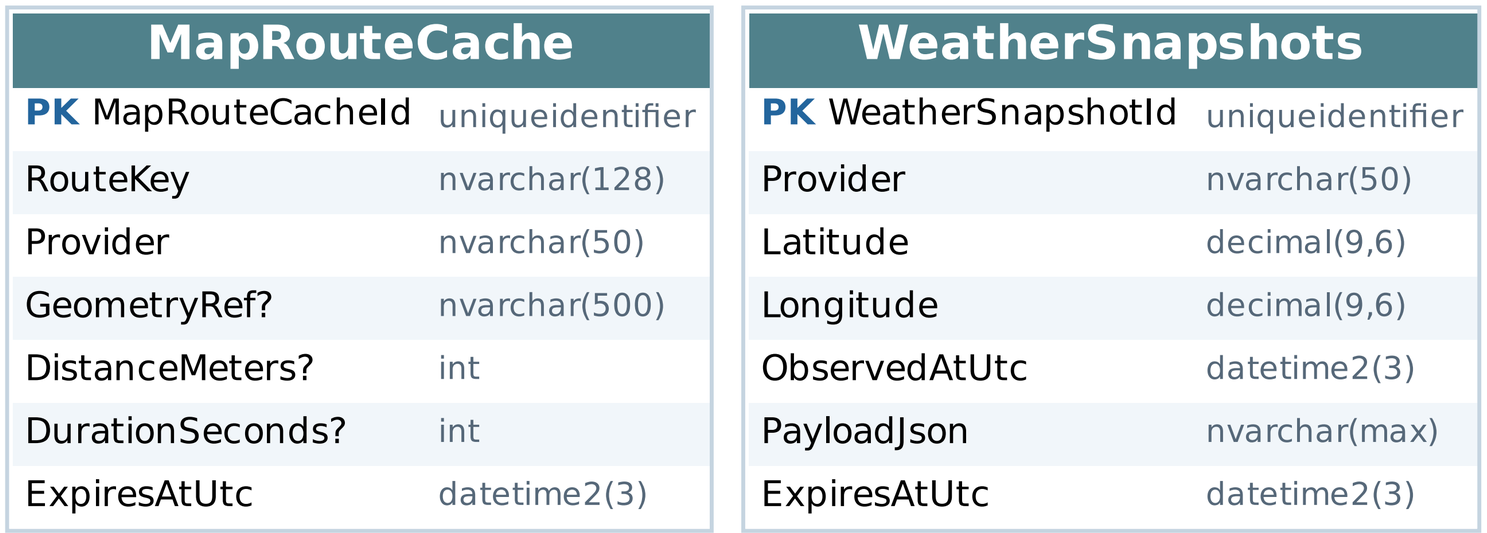
PK OutboxMessageId	uniqueidentifier
EventType	nvarchar(100)
AggregateType	nvarchar(50)
AggregateId	uniqueidentifier
PayloadJson	nvarchar(max)
DispatchStatus	nvarchar(24)
RetryCount	int
PublishedAtUtc?	datetime2(3)
IdempotencyKey	nvarchar(128)

12 / ADMINOPS - 14 TABLES

Admin RBAC, approval queues, audit, rules, reporting | FK arrows: child N -> parent 1




MediaAssets
PK MediaAssetId uniqueidentifier
FK OwnerUserId uniqueidentifier
CloudinaryPublicId nvarchar(400)
ResourceType nvarchar(25)
Visibility nvarchar(24)
UploadStatus nvarchar(24)
ByteSize? bigint
CapturedAtUtc? datetime2(3)
MediaAccessEvents
PK MediaAccessEventId uniqueidentifier
FK ActorUserId uniqueidentifier
FK MediaAssetId uniqueidentifier
AccessType nvarchar(30)
AccessedAtUtc datetime2(3)
Purpose? nvarchar(150)
Memories
PK MemorieId uniqueidentifier
FK MediaAssetId uniqueidentifier
FK PlaceId? uniqueidentifier
FK TripId? uniqueidentifier
FK OwnerUserId uniqueidentifier
Caption? nvarchar(2000)
Visibility nvarchar(20)
MemoryStatus nvarchar(24)


MapRouteCache
PK MapRouteCacheId uniqueidentifier
RouteKey nvarchar(128)
Provider nvarchar(50)
GeometryRef? nvarchar(500)
DistanceMeters? int
DurationSeconds? int
ExpiresAtUtc datetime2(3)
WeatherSnapshots
PK WeatherSnapshotId uniqueidentifier
Provider nvarchar(50)
Latitude decimal(9,6)
Longitude decimal(9,6)
ObservedAtUtc datetime2(3)
PayloadJson nvarchar(max)
ExpiresAtUtc datetime2(3)

CRITICAL DATA RULES & RELATIONSHIPS

Approval, privacy, data integrity and Coins rules that SQL + Backend must enforce

IDENTITY & CONSENT

- One immutable server-issued UserId for Traveler, Hunter and Game; OTP activation and one +100 welcome credit per eligible account.
- Admin access is Users + active AdminUserRoles + fine-grained permissions; no duplicate standalone password table.
- Followers / FollowRequests / TripInvitations / GPS sharing consents are independent relations; blocked profiles restrict access.

TRIP / MAP / SAFETY

- Trip submission is not a completed journey. TripSession requires real StartAt / FinishAt; only valid completions count.
- Real road geometry / ETA comes from provider results, not invented straight-line paths; cache is optional.
- Hazards begin as unverified community reports. Matching alert requires active trip, valid time and route relevance.

HUNTER & GAME APPROVAL

- All new Hunter submissions and Game creations enter Admin queue with UserId + name snapshot + submission ID.
- Hunter live 360-style evidence and photo only; 200-meter duplicate candidate check is not an automatic rejection.
- Game creator hint image can use gallery, player proof must use live camera. Completion is a separate Admin review.
- Public game expires after one month, archived creator record retained. Winner/expiry edge cases await approval.

COINS & VOUCHERS

- Only Coins: +100 welcome, +200 approved hunt, +100 published game, +100 approved eligible completion.
- PostedBalanceCoins is unchanged during Pending redemption; ReservedCoins blocks spending same balance twice.
- Approval finalizes debit once; rejection/cancellation releases hold; voucher QR/code only after real issue.
- Coins award sources uniquely scoped to recipient and source; ledger append-only; transaction retries idempotent.

OPERATIONS & SQL INTEGRITY

- All 17 Admin modules map to one shared database; controller permissions and audit are authoritative.
- FKs, filtered unique indexes and CHECK constraints complement EF transactions, not replace server policy checks.
- Sensitive media is Cloudinary data with SQL ownership metadata; do not expose raw Hunter/Game evidence URLs.
- Do not store Google Maps provider secrets, FCM credentials, OTP codes or real voucher codes as public plaintext.

PENDING OWNER DECISIONS

These are unresolved in the BRD; conditional storage is not an approved live feature

Source documents explicitly leave these decisions open. Do not present them as finalized business rules.

01 **CASH PAYOUT**

Bank/partner payout method, KYC, legal/tax, fees not finalized; PayoutTransactions is conditional only.

02 **REDEMPTION MINIMUM**

Exactly 10,000 Coins or >10,000 Coins requires owner sign-off.

03 **GAME WINNERS**

First-winner only vs multiple eligible players earning 100 Coins remains unresolved.

04 **GAME EXPIRY**

Calendar-month vs 30-day clock anchor and in-flight attempts on expiry need decision.

05 **ADMIN RBAC MATRIX**

17 modules stated, but exact role-to-permission grants and maker-checker requirements not finalized.

06 **REVIEW WORKFLOWS**

Approval turnaround SLAs, escalation, appeal and timeout policies not fixed.

07 **GPS & EVIDENCE**

Media retention, GPS sampling, restricted polygon source and consent-retention schedule to approve.

08 **HAZARD PROXIMITY**

Radius, recency, route bearing and cooldown are configurable but not assigned production values.

09 **OPERATIONAL THRESHOLDS**

Rate limits, upload size caps, data retention and backup RPO/RTO need measurable release values.

Business logic decisions must precede final SQL migration and Admin permissions sign-off.

SOURCES & PROVENANCE

Which parts came from the uploaded specifications and which are physical design proposals

MASTER BRD

Wayhunter_Master_BRD_Revised_Tamil_v2 (1)(2).pdf

24 pages: pp.4–14 user/features/Coins; pp.15–17 Admin 17 modules; pp.18–19 entities; pp.23–24 decisions.

BACKEND DESIGN

Wayhunter_Backend_Architecture(1).pdf

16 pages: pp.3–7 aggregates/project folders; p.10 all 17 Admin modules; pp.13–14 FK, ledger, audit constraints.

FLUTTER DESIGN

Wayhunter_Flutter_Frontend_Architecture_v2_GitHub_BRD_Maps (1).pdf

13 pages: pp.2–10 Screens, maps, feature responsibilities and API integration.

DESIGN NOTE

A full source-derived physical ERD does not exist in the uploaded documents. This package translates source requirements into a proposed relational schema.

Optional structures (e.g. payout and cache) are clearly named conditional, not commitments to a cash payout or an external API contract.

Open data privacy/retention, role grant and SLAs require review before a production database is deployed.